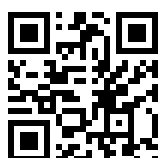


Stablecoins and digital euro: friends or foes of European monetary policy?



Stablecoins and digital euro: friends or foes of European monetary policy?

Abstract

In this paper, we analyse potential impacts that dollar-denominated stablecoins could have on monetary policy. We also consider related questions about the introduction of the digital euro. Based on our findings, the adoption of stablecoins in Europe is unlikely to be large-scale without state support. The development of the digital euro with full state and central bank backing is more likely to succeed. To be a viable alternative to private payment options, the digital euro needs to be carefully designed.

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LIST OF ABBREVIATIONS

ART	Asset-referenced Token
CBDC	Central Bank Digital Currency
DFR	Deposit Facility Rate
DLT	Distributed Ledger Technology
ECB	European Central Bank
EMT	E-Money Token
EU	European Union
EUR	Euro
GENIUS	Guiding and Establishing National Innovation for US Stablecoins
MiCA(R)	Markets in Crypto Assets Regulation
MLF	Marginal Lending Facility
MRO	Main Refinancing Operations
OMO	Open Market Operations
STABLE	Stablecoin Transparency and Accountability for a Better Ledger Economy
US	United States
USD	US dollar

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EXECUTIVE SUMMARY

- **The crypto-asset market exploded in size since the inception of Bitcoin in 2008 and now has multi-trillion valuations.** An increasing number of consumers globally use crypto to trade and speculate. Even some large financial institutions invest in crypto as a risk diversification strategy. Stablecoins were developed as an alternative to the price volatility of unbacked crypto assets. These tokens purport to hold their value 1:1 with a fiat currency. Presently this is mostly the US dollar. Stablecoins are currently used as a bridge between the real economy and the crypto-markets.
- **The current US administration appears to be very favourable to the crypto industry.** Several bills are currently in the process that regulates the industry, intending to spur innovation, protect consumers, and promote stablecoins as a means of payment. Some politicians explicitly call for US-based stablecoins as a tool to expand the US dollar's role as a global reserve currency.
- **The European Central Bank (ECB) has expressed concern around this development.** The dominance of dollar-backed stablecoins in Europe and their potential impact on monetary policy and monetary sovereignty. The ECB also offered this as another argument in issuing the digital euro.
- **In this paper, we investigate to what extent foreign stablecoins could impact the ECB's monetary policy conduct.** We look at the traditional transmission channels and describe how a high stablecoin uptake could have an effect on them. We find that foreign stablecoins are unlikely to witness large adoption in the euro area. By extension, we do not envision that monetary policy will be negatively impacted. The euro is a stable currency, with a broad support of the European public. European payment systems are advanced, inexpensive, and reliable. Europe has already developed the Markets in Crypto Assets (MiCA) regulation to limit large stablecoin issuance. While not perfect, MiCA makes a first step towards stringent crypto-asset regulation. Stablecoins, like other crypto, is related to crime, political corruption, and instability in value. All of these are potential risks to their business model and prone to sudden loss of trust by consumers.
- **The digital euro, being issued by the central bank and denominated in euro, can be a credible alternative to stablecoins in the euro area.** The present vision of the digital euro threatens to fall short of its potential. There are unresolved questions around privacy, competition with banks, and pricing for merchants and consumers. Crucially, the digital euro is currently foreseen to have rather low holding limits. This would severely reduce its usefulness and make it more difficult to compete with other private payment instruments. High holding limits might also have co-benefits for the stability of the banking sector.
- **Stablecoins as a technology are developing rapidly. The US public policy seems to support this growth.** Constant monitoring of the developments in these spaces is merited. Presently, we conclude there is no significant impact of foreign stablecoins on monetary policy in Europe.

1. INTRODUCTION

Crypto-assets are booming. The total market capitalisation is above EUR 3 trillion, with Bitcoin alone at EUR 2 trillion (CoinMarketCap, 2025c). Crypto now also has political support from the United States (US) administration and policymakers. There at least two bills intended to regulate stablecoins currently in the US Senate, while the administration officials are pushing dollar-based stablecoins as a means of supporting the role of US dollar globally (Matthews & Tran, 2025).

European Central Bank (ECB) officials have drawn attention to the possible impacts of these stablecoins might have (Munster & Faggionato, 2025). US-based big tech companies with already significant presence in payment systems could expand their market power by issuing their own stablecoins. This expansion could, according to the ECB, come at the expense of the use of cash and bank deposits as a means of payment in the euro area, but also ultimately impact monetary policy and sovereignty in the euro area. Partly as a result of this growing concern with US stablecoins, the ECB has once again underscored the need for the digital euro as a possible counterweight (Lane, 2025).

In this paper we investigate these concerns. We outline the main transmission channels of monetary policy and discuss how stablecoins could theoretically impact these. We ultimately find that the impact would be negligible as there are severe barriers to their adoption. Euro as a currency enjoys a great support of the Euro are population, bringing into question the need for introducing another currency as a means of payment. The digital payments are already well developed in Europe, leaving little need for stablecoins. European regulation is also very stringent on stablecoin issuers and could limit any large token issuance. Moreover, we find that the instability of stablecoins and the relationship of its issuers with politics and crime might be a risk to the larger adoption. A properly designed digital euro could present another large barrier.

At the same time, the crypto technology and the regulatory space are moving fast, so attention to them is merited. Stablecoins could over time get more support by the US government and the Federal Reserve (Fed), giving them legitimacy and boosting their growth. They could also become more bank-like in nature and get more integrated with the traditional financial sector. This interconnectedness could be a source of contagion and lead to financial instability. As seen in previous financial crises, these could easily have effects on the European financial sector, too. Any response should involve both the regulators and the ECB as monetary policy makers and financial supervisors.

We proceed as follows. In Section 2 we outline design features of crypto assets and the digital euro, comparing them to cash and bank deposits, in order to gauge their potential uptake. Section 3 charts monetary policy transmission channels and how crypto assets could impact them. Section 4 details barriers to adoption of crypto in Euro area. The last section concludes and offers recommendations for policymakers.

2. CRYPTO-ASSETS AND OTHER FORMS OF MONEY

In order to come to a proper assessment of the potential effects of cryptocurrencies on the European monetary system, we need to establish the role they play compared to other forms of money. In this section we discuss five payment instruments: cash, bank deposits, central bank digital currencies (CBDCs), unbacked crypto assets and stablecoins. We outline some of the more relevant features of these instruments and estimate how well they (might) play the role of money in the economy. This clarifies the scope for adoption and possible impacts of crypto assets and CBDCs. Without going into deeper questions on the nature of money, we confine ourselves to discussing the three traditional roles any instrument has in able to perform the role of money well. These are means of payment (being used in transactions for other goods and services); unit of account (are there goods and services invoiced in this instrument); and store of value (does the instrument hold value over time).

2.1. Established instruments: cash and bank deposits

The two most widely used instruments for everyday transactions are cash and bank deposits. Cash is a liability of the central bank and is thus a public form of money (See Table 1 below for full comparison). On the other hand, bank deposits are a liability of the commercial bank issuing it, so it is a private form of money. However, the value of the bank deposits is backstopped by public institutions. Deposit guarantee systems insure deposits up to a pre-determined limit, reinforcing the trust in banks and preventing destabilizing runs. Central banks can also act like lenders of last resort to prevent bank failure and stabilise the value of its liabilities. In some countries in Europe, tax payments are only possible using bank deposits, giving them further legitimacy by the state.

Both cash and bank deposits play the three traditional roles of money well. Being denominated in domestic currency they are a good unit of account. Since they have central bank support they are stable in value (typically in developed countries), thus a good store of value. They also have broad acceptance and are widely accepted means of payment.

2.2. Aspiring instruments: unbacked crypto and stablecoins

In recent years cryptos and stablecoins were developed as a competition to traditional payment instruments. One way to categorise crypto-assets is between *unbacked crypto-assets* (like bitcoin and ethereum) and those that are backed by some kind of asset (like USDT or USDC), often called *stablecoins*. As the value of unbacked crypto-assets does not depend on the value of any other currency or commodity or mechanism, the market value is uncertain and tends to fluctuate considerably over time. This feature partly explains why they are currently more frequently used as a speculation product.

Conversely, the value of stablecoins is kept stable (hence the name) through asset backing. Stablecoin issuers hold liquid financial assets, like cash and government bonds.¹ They can issue one token for each unit of currency (currently usually USD 1) they hold. The tokens they issue are then used for transactions, usually in the crypto markets, and can be later redeemed for fiat currency (DNB, 2022).

When looking at the roles of money, crypto-assets are forms of digital and private means of payment and therefore fulfil the first role of money. Being mined on a DLT, unbacked crypto-assets are not a liability of any legal entity. In contrast, stablecoins have issuers, and the tokens they issue are backed by other assets. However, unlike bank deposits, they currently do not enjoy the level of government support in the form of the deposit guarantee system or lender of last resort by the central bank.

¹ For simplicity we ignore other kinds of stablecoins, such as those backed by commodities like gold, or algorithmically traded stablecoins.

Unbacked crypto-assets are highly volatile in value (see Figure 1 below), leaving them a poor store of value. The transactions of unbacked crypto-assets can be slower and more expensive than cash and bank deposit payments in developed countries (for comparison see section 4.3). This makes them a relatively poor means of payment. Lastly, crypto-assets are not a unit of account. Few prices (in the legal economy, at least) are denominated in crypto-assets. Stablecoins are relatively stable in value and frequently used in crypto spaces as means of payment. However, these are mostly payments for buying and selling other crypto-assets. However, there is very little use of stablecoin in everyday transactions. In summary, crypto-assets do not fulfil all of the functions of money as well as bank deposits and cash do.

Figure 1. Market cap of Bitcoin and Ethereum



Source: CoinMarketCap (2025a, 2025b), authors' own elaboration.

Note: USD billion.

2.3. Proposed instruments: Central Bank Digital Currencies

Central bank digital currencies (CBDCs) in theory have the potential of fulfilling all functions of money. However, CBDCs are still in the research phase for most major economies, including the euro area (Atlantic Council, 2023). As such, design and implementation details are still not finalised. As the name suggests, CBDC's would be digital in nature and issued by the central bank. If CBDCs are issued and transferred on a blockchain the technological aspects will be similar to crypto (ECB, 2024b). However,

it is not clear what benefits this setup would bring for retail² transactions over the traditional, non-DLT based payments, which have broad adoption and high efficiency in Europe (ECB, 2024a).

For CBDC's to effectively fulfil all roles of money, the design choices are important. Unlike bank deposits and stable coins, CBDCs would be liabilities of the central bank and therefore default risk-free and public, making them a potential unit of account and reliable store of value. As for Europe a CBDC would be linked to the euro, it would also function as a unit of account. To make CBDC a means of payment, acceptance and use by the broader public is critical and will depend on design choices made at the introduction. The next paragraph will give insight in the possibilities.

2.3.1. Digital euro

The digital euro is the ECB's CBDC project that is currently in development (ECB, 2023a). One of the main motivators for its introduction is the fact that digital payments using private forms of money are gaining dominance over cash payments (ECB, 2024a). In Europe, these payments are facilitated mostly by US-based Visa and Mastercard payment cards issued by European commercial banks. This presents a two-fold problem for the ECB: fewer Euro area citizens rely on cash as the only public form of money and the reliance on Visa and Mastercard keeps increasing (Cipollone, 2025). This reliance presents a geopolitical risk, as 13 countries in Europe fully depend on non-European card providers and there are only 9 national card schemes remaining in Europe (ECB, 2025d). Moreover, the ECB's concern is that a reduced use of cash would reduce trust in the financial system and bring into question the convertibility of bank deposits (Lane, 2025).

The developments around stablecoins have only increased these concerns as they are by nature private and most of them are expected to be foreign-denominated (Lane, 2025). The ECB allegedly requested revision in the MiCA regulation to further prevent their uptake in the euro area (ECB, 2025a). Moreover, the ECB put forward that the digital euro could help counter these trends. The digital euro would be a public, digital alternative to private and foreign payment instruments.

2.3.2. How the digital euro is envisioned to help

ECB officials put forward several reasons why the digital euro could be an attractive alternative to foreign stablecoins and private means of payment. Firstly, it could be designed to offer a base level of services on top of which various payment providers could build their value-added services. This is opposite of global stablecoins which are foreseen to be locked in within the proprietary platform of its issuer (Lane, 2025). Secondly, the digital euro could be designed to be more protective of the user privacy than current private payment solutions, such that only a minimum of user and payment information is available to the ECB and payment providers (Daman, 2024). The ECB also does not have a profit motive and would not sell this data on to the market (Daman, 2024). Big tech companies, who might become large issuers of global stable coins in the future, and commercial banks are infamous for their privacy violations and often use and sell their customer data for profit (Boissay et al., 2021). Thirdly, the digital euro could be more accessible to the public than the private solutions. For instance, it would be cheaper for merchants, ensuring broader adoption. Moreover, through public institutions and facilitated access it would lower the threshold for use by marginalised groups (ECB, 2024b).

2.3.3. Ensuring the right design

At the same time, a broad adoption of the digital euro is not guaranteed. CBDC rollouts in other countries are a case in point. In case of Nigeria, the Bahamas and Jamaica, after more than a year, the

² Although Eurosystem is experimenting with DLT on the wholesale side, i.e. with large transactions between institutions. See (ECB, 2025c).

total amount of CBDC compared to total currency in circulation was on average still below 1% (Noll, 2024) and in Nigeria only 0.5% of the population had a digital wallet open (Aurazo et al., 2024; Ree, 2023). While these are developing countries and their use-cases for introducing a CBDC differ from the euro area (Aurazo et al., 2024), one of the key learning points is that the end-users would need a clear value proposition from the CBDC (Koonprasert et al., 2024; Noll, 2024).

Since Europe has well developed digital payments systems, this is not a given. This means that any new competitor in the field has a high threshold for widespread user adoption. This is true of stablecoins, but also potentially the digital euro. There are several design choices the digital euro needs to satisfy in order to increase the likelihood of its adoption. First, the digital euro needs to be cheaper and more convenient than existing commercial bank apps, both for merchants and consumers (Bindseil et al., 2021). On the upside, the ECB intends to keep the fees for consumers and merchants minimal, offering a cost advantage to private payment instruments (ECB, 2023a). At the same time, the ECB envisions private banks and payment service providers administering the digital euro (ECB, 2024b). As consumers in the euro area are familiar with their banking solutions, it is not *a priori* clear how the digital euro will be able to differentiate itself and offer more convenience in the wallet it is competing with.

The same is true of privacy. Polling research has found that European consumers highly value privacy in their payments (ECB, 2021). At the same time, the ECB stated that private payment providers sell their clients' data for profit (Daman, 2024). It is then an open question whether consumers would trust the privacy of the public digital euro administered through a private payment solution.

Crucially, in order to meaningfully compete with bank deposits and stablecoins, the digital euro should also be an adequate store of value. This would only be possible if it has high (or infinite) holding limits. This is in contrast with the current ECB plans where holding limits are still to be calibrated, but are expected to amount to EUR 3,000–4,000 per consumer (Panetta, 2022).

Such limitation might hamstring user adoption and ultimately reduce the chances of success of the digital euro (Chiu & Hofmann, 2022; Hofmann, 2023; Niepelt, 2023). It is conceptually not clear why a central bank would provide a physical currency that can be held in (in principle) unlimited amounts, whereas a digital version of it would not (Niepelt, 2023). Similarly, if the ECB is concerned about the loss of the monetary anchor due to the declining use of public money (cash) (ECB, 2023a; Lane, 2025), it would be expected that a more ambitious use of public digital money would be welcomed.

A similar parallel can be drawn with stablecoins. Simply put, stablecoin holdings do not have a cap. If the ECB's stated ambition is to design a CBDC to counter the use of stablecoins (Lane, 2025), it is difficult to see how a limited digital euro could be a more attractive alternative to stablecoins.

The stated reasoning for the low holding limits is reducing the financial stability risks. The concern is that an unlimited digital euro would exacerbate a financial crisis by facilitating fast deposit outflows to the ECB balance sheet (ECB, 2023a). Moreover, mass and long-term migration of bank deposits to digital euro holdings could, according to this reasoning, reduce cheap (deposit) funding from banks, forcing them to acquire alternative (and typically more expensive) funding, which could have consequences for both financial stability and monetary policy (ECB, 2023a) (see discussion at 3.2 below).

First, as recent episodes of financial instability in the US and Switzerland have shown, digital deposits can quickly be withdrawn and moved to another bank with or without CBDCs and stablecoins (Chiu & Hofmann, 2022; Grünewald, 2023). Second, it is not clear why a central bank would prefer that consumers run from deposits into private stablecoins instead of a public CBDC (Hofmann, 2023). Third, and counterintuitively, a run into an unlimited digital euro could give a central bank more information about liquidity needs of impacted banks, leading to a quicker reaction and shortening of the financial instability episode (Chiu & Hofmann, 2022; Hofmann, 2023). Lastly, the ECB can 'recycle' the newly

acquired central bank reserves back into the banking system in the form of new lending to banks (Brunnermeier & Niepelt, 2019)

A more fundamental question is whether the banking system (in Europe) works well and whether the *status quo* should be maintained. Conceptually, it is not clear in the first place that liquid and short-term deposit financing is an appropriate funding source for typically long-term assets banks often hold. Alternative setups are imaginable where different kinds of institutions could match maturities more appropriately (Niepelt, 2023).

A part of the reason for the maturity and liquidity mismatch is the bank regulatory framework. For instance, post-crisis regulatory wave in Europe has increased compliance costs for banks and limited incentives for riskier real economy lending (Chiu & Hofmann, 2022). A part of the regulatory framework that applies to banks specifically is the deposit guarantee system. These systems are designed to compensate bank customers up to a certain point (usually EUR 100,000 in the Euro area) and refund them in the case of the bank bankruptcy. This safety disincentives customers from moving their deposits to competing non-insured products (such as e.g. MMFs), tilting the market playing field in banks' favour.

Having a digital euro with very high (or no) limits could ameliorate this situation. Customers would for the first time have a viable alternative of storing their wealth digitally, this time with a central bank, where it attracts no credit risk, unlike with a commercial bank. This safety could obviate the need for a deposit guarantee system and other bank regulation designed to maintain deposit safety (Chiu & Hofmann, 2022). Moreover, this could reduce bank privilege, expose banks to a fair(er) market competition and price risk in more accurately, without government guarantees. A more level playing field between banks and other credit providers could lead to new lending opportunities and help serve underfunded sectors, such as productive small and medium enterprise lending (Chiu & Hofmann, 2022)

To sum up, bank deposits and cash are established payment instruments and are very money-like. CBDCs are still not in existence in developed economies and their use as money will largely depend on the design choices. If the holding caps are strictly enforced the store of value role could be limited compared with other money substitutes on the market. Means of exchange and unit of account role also depend on whether the consumers and merchants widely adopt it and see the added value compared to other payment instruments on the market, as the case of Nigeria and Jamaica has shown.

Crypto-assets provide innovative, peer-to-peer payment options, but have not broken into the larger economy, so everyday goods and services cannot be purchased with them. Unbacked crypto assets are a poor store of wealth as their value fluctuates considerably. Stablecoins could be a better store of value, but their stability has not always been guaranteed (see discussion in Section 4) and typically are not remunerated, leaving them disadvantaged to some other instruments, such as bank deposits or MMFs.

Table 1. Properties of different forms of money and crypto-assets.

	Cash	CBDCs	Bank deposits	Unbacked crypto	Stablecoins
Digital	No	Yes	Yes	Yes	Yes
Public	Yes	Yes	No	No	No
On a blockchain	No	Not expected in euro area ³	No	Yes	Yes
Programmable	No	Possible, but digital euro not ⁴	Optionally	Yes	Yes
Use without intermediaries	Yes	Not in Euro area ⁵	No	Yes	Yes
Has large price volatility	No, stability guaranteed by the central bank	No, stability guaranteed by the central bank	No, parity with cash guaranteed by the central bank	Yes	Not typically
Means of payment role	Yes	If broadly adopted	Yes	Not in the wider economy	Not in the wider economy
Unit of account role	Yes, denominated in domestic currency	Yes, denominated in domestic currency	Yes, denominated in domestic currency	Not in the wider economy	Not in the wider economy
Store of value role	Yes	Not if holding limits are low	Yes	No	Yes
Performing money role	Well	Depending on the design	Well	Presently poorly	Presently poorly

Source: Banco de España (2022), authors' own elaboration.

³ See (ECB, 2024b), but wholesale (i.e. inter-institutional) settlements experiments already began (ECB, 2025c).

⁴ See (Panetta, 2023)

⁵ See (ECB, 2023a)

3. INTERACTION BETWEEN MONETARY POLICY AND STABLECOINS

In this section we outline the goals, instruments, and transmission channels of monetary policy in Europe. We discuss mechanisms by which stablecoins could theoretically impact monetary policy in Europe. In Section 4 we analyse the potential impact of stablecoins on these transmission channels. In the following section we show how the digital euro could reduce this impact even further.

The ways in which the monetary policy instruments of the ECB affect the real economy are called transmission channels. On the upstream of this transmission there is the ECB's adjustment of the short-term rate that determines the medium- and long-term rates through expectations. On the downstream is the cost and availability of liquidity for companies and households (Lane, 2022). For this paper we focus on the effect of stablecoins on the most relevant downstream channels.

3.1. Interest rate channels

Interest rate channels refers to the way in which short-term interest rates set by central banks have effects on consumer and company savings, wealth, and cash flow. For instance, higher interest rates make household savings more attractive, reducing consumption in the short term. They also reduce financial asset (stock and bond) prices, decreasing total wealth and reducing spending. Higher interest rates impact the cash flow of consumers and companies: on the one hand it increases interest payments, reducing the cash flow available for consumption; on the other hand, it increases interest revenues of the assets consumers and companies might hold (Lane, 2022).

Multiple studies have found that monetary policy shocks impact crypto-asset prices, although there are differences in terms of the sign and size of the effect, as well as the jurisdiction. For instance, Karau (2023) finds that after 2020 Federal Reserve's contractionary monetary policy (i.e. increase in interest rates) reduced the price of Bitcoin, but that in the pre-2020 period this contractionary stance had the effect of increasing its price. A study by Ma and collaborators contradicts the latter finding. They find that Fed's monetary tightening is associated with a decrease in the price of Bitcoin on a pre-2020 dataset (Ma et al., 2022). International Monetary Fund research finds that an increase in Fed interest rates decreases the market capitalisation of the crypto markets (Che et al., 2023). Another study finds that Fed monetary policy tightening decreases the market capitalization of stablecoins specifically, as higher interest rate environment prompts investors to move from non-yielding stablecoins to traditional investment vehicles (Aldasoro et al., 2025).

In the euro area, a decrease in interest rates is found to reduce Bitcoin prices (Pietrzak, 2023). A Bundesbank study found different results: a decrease in ECB interest rates leads to an increase in Bitcoin prices, although the effect found was quite small (Bundesbank, 2021). Monetary policy then does appear to have an impact on crypto-assets, but the effect sign is not conclusive and a matter of further empirical study.

3.2. Bank lending channels

Bank funding costs, i.e. the price at which banks refinance themselves, are key for monetary policy transmission. Banks typically get their funding either through deposit issuance (i.e. deposit funding), which is shorter term, cheaper, and more 'sticky' (slower to respond to monetary policy impulses); or through wholesale funding, which is usually longer term, more expensive, and quicker to respond to monetary policy. Increase in bank funding prices translates into the price of new lending or existing flexible-rate loans to the economy. An increase in interest rates also reduces the prices of assets that banks hold and accept as collateral, reducing the lending rate and overall consumption (Lane, 2022).

The literature indicates several ways that crypto-assets might impact the bank lending channel. For instance, large scale consumer preference for crypto-assets might lead to deposit flight from commercial banks. This would, the argument goes, force banks to shift from deposit funding to wholesale funding. This could make bank lending more expensive and volatile (Assenmacher, 2020; G7 Working Group on Stablecoins, 2019). At the same time, it could improve the policy transmission, as wholesale funding is more responsive to monetary policy impulses than cheaper but more 'sticky' deposits (Beltrametti & Pittaluga, 2023; G7 Working Group on Stablecoins, 2019). Lending channel could also be weakened if large amounts of bank holdings switch to crypto-assets and away from traditional assets that central banks can directly influence (Melachrinou & Pfister, 2021). *In extremis* stablecoin issuers could become bank-like in their activities, engaging in lending and thus creating new tokens that are not fully backed by assets (as is now the case), and weakening monetary policy transmission (G7 Working Group on Stablecoins, 2019).

There are reasons to doubt any of these impacts could take place in the euro area. Firstly, allowing stablecoin issuers to engage in bank-like activities is a policy choice. Banks are faced with strenuous financial regulation and supervision, and are subject to bank charter laws. Presently, through the MiCA regulation, stablecoin issuers would not be treated like banks and would be subject to strong requirements on which assets they would need to hold to back their coins (Regulation (EU) 2023/2869, 2023).

Secondly, as discussed previously, monetary policy does impact crypto-asset values, although the effect and the sign are not clear. Consequently, even if crypto-assets do become a larger share of bank holdings, monetary policy effect would not by default be weakened, as suggested by Melachrinou and Pfister (2021).

Most importantly, it is far from certain that (European) consumers would *en masse* convert their bank deposits to stablecoin holdings to such an extent that it might impact monetary policy (see discussion in 4.3 below). Nor is it certain that banks would passively accept competition from stablecoins. For instance, findings show that banks have previously increased deposit rates in response to increased competition from fintech companies (Hodula, 2023). This reaction could conceivably be replicated against stablecoin issuers.

3.3. Exchange rate channel

Finally, monetary policy changes impact currency exchange rates: an interest rate decrease by the ECB depreciates the value of the euro relative to other currencies (Lane, 2022). Assuming that, for example, dollar-denominated assets have not changed in yield, they automatically become more attractive than euro assets. This increased demand in dollars and reduced demand in euro leads to depreciation of the latter (Bundesbank, 2021). The depreciation of the currency also makes domestic goods cheaper, causing a rise in exports and, in turn, increased demand (Mishkin, 2019).

As crypto-assets are *de facto* akin to foreign currencies, the same mechanism should apply: reducing interest rates will depreciate the domestic currency vis-à-vis stablecoins (Bundesbank, 2021), leading to an increase in consumption in domestic currency, leading to strengthening in this transmission channel.

This holds under the assumption that foreign currency plays the medium of exchange, but not the unit of account role. If prices in an economy become (partly) denominated in a foreign currency, central bank by definition loses the grip on price stability as prices are not denominated in the assets in can intervene on (barring foreign reserves it might hold). This (partial) repricing of goods in foreign currency is called dollarization. Some authors fear (or at least discuss) this scenario might be replicated

with a successful uptake of foreign-denominated stablecoins (see for example Beltrametti & Pittaluga, 2023; ECB, 2019; He, 2021; Melachrinou & Pfister, 2021; Pfister, 2017; Stevens, 2017)

While it is true that the largest stablecoins are dollar-denominated, it is not a foregone conclusion that these stablecoins would broadly be used in Europe. Moreover, MiCA is especially designed to maintain financial stability and limit the influence of foreign currency-backed stablecoins (Regulation (EU) 2023/2869, 2023)).

However, most importantly, dollarisation is largely associated with developing countries where trust is lost in domestic currency (Pfister, 2017). It is hard to conceive that the euro area, with a strong backing in laws, stable currency and well-functioning payment systems, would need to resort to dollarisation. In a remote scenario where such a threat were to arise, governments have mechanisms to defend their domestic currencies, such as, not allowing tax payments in this currency, enforcing public procurement contracts only in domestic currency, etc (Pfister, 2017; Stevens, 2017). We discuss more below.

4. STABLECOIN ADOPTION IN EUROPE

The above overview shows how theoretically foreign-denominated stablecoins could impact European monetary policy. As there is very little academic literature on these mechanisms our analysis is based on theory rather than empirical evidence of the described channels. However, based on the theory available and the empirical data with regards to the size and use of stablecoins in Europe, there are reasons to doubt this might materialize. There are four important reasons why. It is doubtful to what extent these concerns from the literature impact the European context.

4.1. Institutional barriers

The euro is the official currency in the euro area. It is a legal means of payment in all of the euro area (ECB, 2023b). Euro banknotes are easily recognisable, widely in use, and are an explicitly a symbol of European identity (ECB, 2025b). Retail prices in the euro area are denominated in euro, salaries are paid in this currency, and taxes are levied in it. The euro is also adopted by some non-EU countries due to the strength of the trade with EU and trust in the value of the currency (ECB, 2023b). The ECB maintains the monetary stability of euro and through supervisory action maintains parity between the euro in physical form and private bank deposits (Lane, 2025). Europe on the whole has well developed digital payments systems. Percentages of payments by cards, digital wallets and online payments are high and growing in many member states (ECB, 2024a). In the latest survey on the topic, 83% of the citizens of the Euro area have a positive view of the Euro, the highest score recorded yet (European Union, 2025). The familiarity and trust in a currency supported by the government give the currency considerable stickiness.

This government support has been a strong driving force for millennia. According to historical and anthropological records, states have had a decisive role in defining and circulating currencies within their territories (Goodhart, 1998). They had a crucial role in establishing the unit of account for everyday trade (Goodhart, 1998; Graeber, 2014; Hudson, 2004). Other (private) currencies could still circulate, as long as the state had the last word in setting the terms of the trade (Graeber, 2014; Knapp, 1924).

This view helps understand the success of bank deposits. While a private instrument, they are usually denominated in a domestic currency and the parity with the public currency is maintained through central bank policymaking. But the state-view of money might also explain why foreign-denominated stablecoins might not succeed in Europe: they are neither denominated in domestic currency, nor have an explicit backing and parity commitment by the central bank. The Fed has not committed to acting like the lender of last resort for stablecoin issuers, nor is there any kind of a deposit guarantee system for them. This lack of state support could reduce the trust among the users of the currency and hamper the adoption of foreign-based stablecoins.

Aside from setting the unit of account, the states have historically been the privileged issuers of currency and were uniquely entitled to income (seigniorage) from minting the currency tokens (Huber, 2017). The government has also had a unique prerogative to tax its population. For instance, Confederacy during the US Civil War started issuing new currency to help pay off debt with seigniorage income and used taxation to stimulate the circulation of that currency (Goodhart, 1998). This effect of introducing new currency and taxation was also demonstrated in various colonial endeavours (Goodhart, 1998; Graeber, 2014).

There are clearly exceptions, such as dollarised countries that adopt other sovereign currencies, or a unique example El Salvador that for a period designated Bitcoin as a legal tender. But these examples almost exclusively apply to countries going through social unrest, low trust in government, and economic instability. Other historical examples also demonstrate that the collapse of the currency follows the collapse of the issuer (Goodhart, 1998).

Crucially, none of this is the case with the euro area countries and institutions today. Strong governance, rule of law and trust in domestic currency characterise present situation, and it is doubtful that mere convenience of payment provided by foreign stablecoins could counter these overwhelming historical and political forces.

4.2. Regulatory barriers

The development of Libra (later renamed to Diem), a global stablecoin envisioned by Facebook (since renamed to Meta), in 2019 has prompted the regulatory response within the EU. Hence, the Markets in Crypto Assets (MiCA) Regulation was born (Regulation (EU) 2023/2869, 2023).

MiCA focuses mostly on stablecoin issuance, and not the whole of the crypto market.⁶ Stablecoins are not defined by name, instead broken down into two groups: Asset-Referenced Tokens (ART) and E-money Tokens (EMT). ART stabilise their value referring to another value, right, or one or more currencies. EMT maintain their value by referencing only one currency (Zetzsche & Sinnig, 2025). Both types of tokens need to have a prospectus ('white paper') containing the most important properties of the tokens, issued by a recognised EU legal person, and be backed by assets segregated from the issuer's assets (Cantú et al., 2025). The EMT Regulation mandates that at least 30% of the assets backing EMT are in credit institutions (i.e. banks) and tokens be backed 1:1 with liquid reserves (Cantú et al., 2025; Zetzsche & Sinnig, 2025).

Interestingly, MiCA also entails very strict requirements for larger and more successful ART and EMT. Both fall under special European Banking Authority supervision if they reach more than 10 million users, 2.5 million transactions and EUR 500 million in transaction value per day. More extremely, a cap is placed on an ART with 1 million transactions per day and EUR 200 million total transaction value in one currency area. These issuers will need to stop trading and issue a plan to reduce the trade volume (Zetzsche & Sinnig, 2025).

It is still unclear what net effect MiCA will have on stablecoin issuance and the use of foreign tokens in Europe as its rollout was quite recent. Tether, the issuer of the biggest stablecoin by volume, has already announced they would not try to register in Europe, partly due to the regulatory burden, but also because Europeans are not a large part of their customer base (Peak, 2025). On the other hand, large European banks like ING are reportedly planning to join a consortium of banks and start issuing their own stablecoin (Allison, 2025).

If the ultimate goal of MiCA was to slow down large-scale, foreign stablecoin adoption, based on these initial results it appears the goal has been achieved for now. The question remains whether this regulation will promote issuance of domestic stablecoins (to the extent this is desirable in the first place) and if the capping presents an obstacle to potentially innovating effects of stablecoins in the payments space.

4.3. Credibility of crypto

Stablecoins, like other crypto-assets, come with downsides on their own. These might present an additional threshold to adoption in the euro area. Contrary to their name, stablecoins are not always stable in value. In fact, in several episodes they lost parity with the fiat currency. This means that less than USD 1 could be obtained for one stablecoin token. For instance, USDT lost the parity in 2018 and 2023 (Browne, 2018; Nicolle, 2023). The instability of USDT is related to controversy around Tether, the issuer of USDT, failing to show multiple times in an audit that all of its tokens were backed by liquid

⁶ For instance, it does not regulate Non-Fungible Tokens and unbacked crypto assets, but does put cover Crypto Asset Service Providers, i.e. exchanges.

dollar instruments and alleged fraud involving the Bitfinex trading platform (Browne, 2021; Lopatto, 2021). Similarly, TerraUSD, an algorithmic stablecoin was found to be associated with a crypto-Ponzi scheme in 2022, leading to the loss of trust, withdrawal of reserves among coin holders, and eventual collapse of the stablecoin (DNB, 2022).

The relationship between crypto and crime is broader. One analysis puts the total volume of crypto used in criminal activities at USD 41 billion as a lower bound (Chainalysis, 2025). The acts in question include not just crypto-market investments, like pyramid schemes and fraud, but also more traditional crime operations such as drug trade, money laundering, and child sexual abuse material (Europol, 2021). Crypto is also used for sanctions evasion, with an estimation that state-run North Korean hacking groups managed to steal around USD 1 billion in 2022 alone (Clark et al., 2022; Naumaan, 2023). The true extent of sanctions evasion by crypto-assets is likely to be much larger than this.

Concerns also exist that the US regulation of stablecoins is too lax and might lead to corruption. US president Donald Trump has reportedly already benefitted from his own crypto company World Liberty Financial to the tune of USD 1 billion (Alexander, 2025). The GENIUS act, according to its detractors, does not offer enough guardrails against US politicians benefitting personally from such stablecoin issuance (McNair, 2025). This political involvement could be a political liability, and suddenly shift trust of the token holders and cause them to lose value.

The Bitcoin's original design was enabling cheaper transactions than established institutions (Nakamoto, 2008). In terms of cross-border fees that might be the case, as traditional fees amount to on average globally 6.62% and could take days to settle (World Bank, 2024), whereas crypto-asset fees⁷ in the previous months averaged between USD 1 and 2 for Bitcoin transactions⁸ (BitInfoCharts, 2025) and settle in seconds or minutes. However, this cost advantage of crypto might be of less relevance for Europe, where settlement is typically instantaneous, even for cross-border transfers (European Payments Council, 2025), and card fees for merchant transactions are very low, at 0.2% to 0.3% for Visa and Mastercard payments (Mastercard, 2015; Visa, 2025).

However, transaction fees do not capture the full price of crypto use. For instance, one study found that a single Bitcoin transaction required 402kg of CO₂ emissions, amounting to EUR 37 per transaction (Trespalacios & Dijk, 2021) – much larger than the market price of the transfer fee. The total environmental costs of Bitcoin mining were estimated at EUR 2.4 billion in 2020 (Trespalacios & Dijk, 2021). These amounts are certain to be larger currently, as the energy required has doubled from 70TWh to at least 150TWh in 2024 (Delahaye, 2024). These findings bring into question the alleged price benefits of stablecoins. End users might not find these externalities a relevant factor in their choice of payments. However, governments, especially those with strong climate and nature protection commitments, would be advised to take these considerations into account when regulating and legitimating the crypto sector.

Presently, crypto-assets are not predominantly used as means of payment. For instance, in Europe only 9% of consumers held crypto in 2024, and of those only 14% held them only for payment purposes. On the other hand, on average two thirds of Europeans who hold crypto stated they used them for investments (ECB, 2024a). The original vision of Bitcoin as a decentralized and disintermediated means of payment is replaced by a reality of crypto-assets often being used as investment and speculation vehicles (DNB, 2022). Broader adoption of crypto-assets for speculation might have financial stability effects, especially if more large financial institutions start investing in crypto. However, for now the

⁷ Crypto-asset transfer fees are flat, market-based and depend on congestion (i.e. demand) at the time of the desired transfer (DNB, 2022).

⁸ But have historically run into dozens or hundreds USD per transaction (BitInfoCharts, 2025).

baseline retail stablecoin use as a means of payment is very low, especially outside the crypto-markets. It is not clear what their trajectory for growth will be.

4.4. What to monitor and how to react

The above analysis considered the *status quo* of stablecoin development and concludes that, at the current pace, there is little reason to worry that they could seriously impede euro area monetary sovereignty and monetary policy. This does not mean European policymakers should be complacent, but carefully monitor developments in this space.

Stablecoins could provide such innovation that they gain broad adoption with consumers and merchants. They might prove to be cheaper to transfer, leading merchants to provide discounts on stablecoin payments compared to the more traditional means of payment. Large supermarket chains, tech, media providers could use their size to quickly scale up this demand. Stablecoin issuers might also engage in (unregulated) credit provision, catering to customers underserved by traditional lenders. All of these trends could suddenly materialise in the moment of crisis and lack of trust in the financial sector.

European regulators should follow this trend carefully. Updates to MiCA could be warranted, especially in a quickly innovating environment. At the same time, regulation should not be so strict as to stifle genuine innovation. Bank business models should not necessarily be protected at all costs, especially if that would hamper customer benefits. Moreover, the digital euro could prove to be cheaper for merchants than private payment solutions, making it a viable alternative to stablecoins. An appropriate set of design choices could guarantee this.

Stablecoins could also gain more government support, especially in the US. The GENIUS Act is quickly progressing through the US Senate. The supporters of the bill promise more innovation and customer choice in the stablecoin space (McNair, 2025). If that materialises, stablecoins could become even more attractive across the Atlantic. The US government could also introduce consumer protection rules similar to banking rules presently. For instance, deposit guarantee scheme could be extended to cover bank stablecoin issuance. Stablecoin issuers could be allowed to lend out crypto, mimicking banks even further. Big banks could migrate to the DLT on the backend, using stablecoins for interbank trade instead of central bank reserves. This could force Fed to acquire and hold stablecoins, using them for monetary policy increasingly in place of the US dollar (Melachrinou & Pfister, 2021). More extremely, the Fed might be pushed to hold crypto assets as a part of its foreign exchange portfolio, as already hinted at by Trump with his Strategic Bitcoin Reserve plan (Chavez-Dreyfuss & Mattackal, 2025; Hunnicutt, 2025).

These trends could introduce instability in the US financial sector, such as deposit shifts to stablecoins in the case of bank crisis. As in 2008 and 2020 crises, this might have effects on the stability of the banking sector in the euro area. While not immediately stablecoin related, this is something bank supervisors should continuously monitor.

While this US administration has already tested the limits of Fed's independence, it is not a given that Fed will adopt crypto in its balance sheet easily (Derby & Saphir, 2024). Moreover, stablecoins might weaken Fed's monetary policy and ultimately the role of the dollar as a global reserve asset. This is contrary to the administration's ambitions, who reportedly intend to use stablecoins to promote the use of the dollar internationally instead to weaken it (Matthews & Tran, 2025). The ECB has in any case excluded holding crypto as a part of its portfolio (Reuters, 2025). Moreover, the EU might take a page out of US's book and use euro-denominated stablecoins to promote the role of the euro internationally. The ECB has long had this ambition (van 't Klooster, 2023), which only intensified in the last several months (Lagarde, 2025).

5. CONCLUSIONS AND RECOMMENDATIONS

The current US administration is developing regulation to promote the market for stablecoins. According to one interpretation, stablecoins could then be used as tools of statecraft and strengthen the global reserve currency role of the US dollar. The ECB officials have shared concerns that foreign stablecoins could have a negative effect on monetary sovereignty and monetary policy.

This study looks at the monetary policy transmission channels in Europe investigates what the theoretical impacts of stablecoins could be. Our analysis finds that at this point in time, significant impact is unlikely. We find that there are significant institutional and regulatory barriers to wider adoption of foreign stablecoins in the euro area. The euro as a currency enjoys remarkable trust across the euro area. Europe also has well developed, stable, and fast digital payment systems. These factors bring into question the motivation of Europeans to switch to a new means of payment.

The MiCA regulation entails stringent criteria for stablecoin issuance in Europe, as well as rigorous provisions for disciplining issuers with large amounts of tokens in circulation. This has already discouraged large foreign issuers from registering in Europe. Stablecoins are often used in various kinds of crime, including fraud, extortion, sanctions evasion and corruption. Their issuers were involved in criminal activities, often leading to a loss in the value of the token. These links with crime and news of instability could lead to a loss of trust in stablecoins, causing broader financial instability.

The digital euro is ECB's flagship CBDC project, with one of the goals of offering a credible alternative to stablecoins. Being issued by the central bank and euro-denominated, it might prove to be familiar to consumers and enjoy wide adoption. This crucially depends on key design choices. As it now stands, the digital euro might prove to be underwhelming to European consumers. Privacy, price, and added value compared to other private payment solutions might put them back from the digital euro. Crucially, the holding limits need to be increased considerably for the digital euro to be a viable payment alternative to cash, bank deposits, and stablecoins.

Supervisors, regulators and monetary policy makers should carefully monitor the developments in the crypto spaces and the US policy related to stablecoins. Given little regulatory pressure, stablecoins might in time become more like banks, potentially leading to financial instability that could impact European banking sector. The US government and the Fed might give stablecoins a bigger role, making it a staple of banking and monetary policy. These could have an impact on ECB's financial and monetary policy making.

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In this paper we analyse potential impacts that dollar-denominated stablecoins could have on monetary policy. We also consider related questions about the introduction of the digital euro. Based on our findings, the adoption of stablecoins in Europe is unlikely to be large-scale without state support. The development of the digital euro with full state and central bank backing is more likely to succeed. To be a viable alternative to private payment options, the digital euro need to be carefully designed.

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